

OBJECT-ORIENTED SOFTWARE ENGINEERING

UNIT 02 : PROJECT MANAGEMENT, PLANNING AND RISK ANALYSIS

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UNIT 02: PROJECT MANAGEMENT

COCOMO

Partial reference: COCOMO-I, Classroom Presentation, Jagat Man Mulguthi, NCIT, Spring 2018

SOFTWARE COST ESTIMATION METHODS

- Cost estimation: prediction of both the person-effort and elapsed time of a project
- Methods:
  - Algorithmic
  - Expert judgment
  - Estimation by analogy
  - Parkinsonian
  - Price-to-win
  - Top-down
  - Bottom-up
- Best approach is a combination of methods
  - compare and iterate estimates, reconcile differences
- COCOMO - the "COnstructive COst MOdel"
  - COCOMO II is the update to Dr. Barry Boehm's COCOMO 1981
- COCOMO is the most widely used, thoroughly documented and calibrated cost model

INTRODUCTION

- proposed by Barrey W. Boehm in 1981.
- Empirical software estimation model in the world.
- Predicts the **effort** and **duration** of a project based on :
  - inputs relating to the size of the resulting systems and
  - number of "**cost drives**" that affect productivity.

THE DEVELOPMENT MODE

1. **Organic mode** :
    - **Relatively simple** ,
    - **small projects** with a **small team** are handled .
    - **good application experience** to **less rigid requirements**.
2. **Semidetached mode**:
    - For intermediate software **projects(little complex** compared to organic mode projects in terms of size).
    - have a **mix of rigid** and less than **rigid requirements**.

3.Embedded mode:

- Project to be developed within a **tight set of hardware and software operational constraints**.
- Example of complex project: Air traffic control system

- The coefficients of A_a, b_b, c_c, d_d for the three modes are:

	A_a	b_b	c_c	d_d
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

COCOMO: SOME ASSUMPTIONS

- Primary cost driver is the number of Delivered Source Instructions (DSI) / Delivered Line Of Code developed by the project
- COCOMO estimates assume that the project will enjoy good management by both the developer and the customer
- Assumes the requirements specification is not substantially changed after the plans and requirements phase.

FORMS OF COCOMO MODELS

- COCOMO is defined in terms of three different models:
  1. **Basic model**,
  2. **Intermediate model**, and
  3. **Detailed model**.

1. Basic Model:
   - a **function of program size** expressed in terms of **lines of Code (LOC)**.
   - **good for quick, early, rough order of magnitude estimates of software costs** .

1.BASIC COCOMO

- a **function of program size** expressed in terms of **lines of Code (LOC)**.
- **good for quick, early, rough order of magnitude estimates of software costs**
- It does not account for differences in
  - hardware constraints,
  - personnel quality and experience,
    - use of modern tools and techniques, and
    - other project attributes

1.BASIC COCOMO

- The basic COCOMO model takes the following form:

  $E=a_b$ (KLOC)^{bb} persons-months

  $D=c_b$ (E)^{db} months

  where,

  E - Stands for the effort applied in terms of person months

  D - Development time in chronological months

  KLOC - Kilo lines of code of the project.

  From E & D we can compute the no. of people required to accomplish the project as

  $$N=E/D$$

1.BASIC COCOMO

- **Merits**
  - good for **quick**, **early**, **rough order of magnitude estimates of software project.**
- **Limitations**
  - **Limited Accuracy:** does not consider certain factors (**hardware constraints**, **personal quality** and **experiences, modern techniques and tools**) for **cost estimation** of software.
  - [redacted]

1.BASIC COCOMO

Example: Consider your team is developing a software project using semi-detached mode with 30,000 lines of code . We will obtain estimation for this project as follows:

(1)Effort estimation

(2) Duration estimation

(3)Person estimation

1.BASIC COCOMO

Example: Consider your team is developing a software project using semi-detached mode with 30,000 lines of code . We will obtain estimation for this project as follows:

(1)Effort estimation

$E = a\,(KLOC)^{b}$ person-months

$E = 3.0(30)^{1.12} = 135.36$ (**136**)

(2) Duration estimation

$D = c\,(E)^{d}$ months

$D = 2.5\,(136)^{0.35} = 13.95$ (**14**)

(3)Person estimation

$N = E/D$ persons

$N = 136/14 = 9.7$ (**10**)

2. INTERMEDIATE COCOMO

- Computes effort as **a function of program size** and a **lot of cost drivers** that includes subjective assessment of
  - **product attributes**,
  - **hardware attributes,**
  - **personal attributes** and
  - **project attributes**

	Cost Driver	COCOMO_II Post Architecture	COCOMO_II Early Design	REVIC	COCOMO_87	COCOMO_81 & COCOMO_85
Personnel Factors						
ACAP	Analyst Capability	Yes		Yes	Yes	Yes
APEX AEXP	Applications Experience	Yes		Yes	Yes	Yes
PCAP	Programmer Capability	Yes		Yes	Yes	Yes
LEXP	Programming Language Experience			Yes	Yes	Yes
VEXP	Virtual Machine Experience			Yes	Yes	Yes
PERS	Personnel Capability		Yes			
PREX	Personnel Experience		Yes			
PCON	Personnel Continuity	Yes				
PLEX PEXP	Platform Experience	Yes				
LTEX	Language and Tool Experience	Yes				
Product Factors						
RELY	Required Software Reliability	Yes		Yes	Yes	Yes
DATA	Database Size	Yes		Yes	Yes	Yes
CPLX	Software Product Complexity	Yes		Yes	Yes	Yes
RUSE	Required Reusability	Yes	Yes	Yes	Yes	
DOCU	Documentation Match to Life-Cycle Needs	Yes				
RCPX	Product Reliability and Complexity		Yes			

	Cost Driver	COCOMO_II Post Architecture	COCOMO_II Early Design	REVIC	COCOMO_87	COCOMO_81 & COCOMO_85
Platform Factors						
TIME	Execution Time Constraint	Yes		Yes	Yes	Yes
STOR	Main Storage Constraint	Yes		Yes	Yes	Yes
TURN	Computer Turnaround Time			Yes	Yes	Yes
VIRT	Virtual Machine Volatility			Yes		Yes
VMVH	Virtual Machine Volatility: Host				Yes	
VMVT	Virtual Machine Volatility: Target				Yes	
PVOL	Platform Volatility	Yes				
PDIF	Platform Difficulty		Yes			
PLAT	Platform			Yes		
Project Factors						
TOOL	Use of Software Tools	Yes		Yes	Yes	Yes
MODP	Modern Programming Practices			Yes	Yes	Yes
SCED	Required Development Schedule	Yes	Yes	Yes	Yes	Yes
SECU	Classified Security Application			Yes	Yes	
SITE	Multisite Development	Yes				
FCIL	Facilities		Yes			
RVOL	Requirements Volatility			Yes		

2. INTERMEDIATE COCOMO

- Based on the rating effort multipliers is determined. The product of all effort Multipliers result in “effort adjustment factor” (EAF). The intermediate COCOMO takes the form.
- $E = a_i (KLOC)^{bi} \times EAF$ where

  E - Effort applied in terms of person-months

  KLOC - Kilo lines of code for the project

  EAF - It is the effort adjustment factor
- The duration and person estimate is same as in basic COCOMO model i.e;

  $D = c_b (E)^{db}$ months i.e; use values of c_b and d_b coefficients.
- N=E/D persons

2. INTERMEDIATE COCOMO

Software Projects	A_i	B_i
Organic	3.2	1.05
Semi-detached	3.0	1.12
Embedded	2.8	1.20

2. INTERMEDIATE COCOMO

- **Merits**:
  - This model can be applied to almost entire software product for easy and rough cost estimation during early stage.
  - It can also be applied at the software product component level for obtaining more accurate cost estimation.

- **Limitations:**
  - The effort multipliers are not dependent on phases.
  - A product with many components is difficult to estimate.

2. INTERMEDIATE COCOMO

Example: Consider your team is developing a project having 30,000 lines of code which in an embedded software with critical area hence reliability is high .

The estimation can be

$E = a\,(KLOC)^{bi} \times EAF$

As reliability is high

EAF=1.15 (product attribute)

$a_i =$

$b_i =$

$E =$

$D = c_b\,(E)^{db}$

$=$

$N = E/D$

2. INTERMEDIATE COCOMO

Example: Consider your team is developing a project having 30,000 lines of code which in an embedded software with critical area hence reliability is high .

The estimation can be

$$E = a\,(KLOC)^{bi} \times EAF$$

As reliability is high

EAF=1.15 (product attribute)

a_i =2.8

b_i =1.20 for embedded software

$$E = 2.8\,(30)^{1.20} \times 1.15$$

$$= 191 \text{ person month}$$

$$D = c_b\,(E)^{db}$$

$$= 2.5(191)^{0.32}$$

$$= 13 \text{ months approximately}$$

$$N = E/D$$

$$= 191/13$$

N = 15 persons approx.

3. DETAILED COCOMO MODEL

- Detailed COCOMO incorporates all characteristics of the intermediate version with an assessment of the cost driver's impact on each phase of the development process.
- This helps in determining the man power allocation for each phase of the project.

3. DETAILED COCOMO MODEL

- **Phase-Sensitive Effort Multipliers**:
  - Some phases(Design, programming, integration/test)are more affected than others by factors defined by the cost drivers.
- **Three –Level Product Hierarchy:**
  - Modules,
  - Sub-System, and
  - System Levels
- The rating of the cost drivers are done at appropriate level. i.e. the level at which it is most suspicious to variation.

3. DETAILED COCOMO MODEL

- Phases
  - Plans and requirements
  - System Design
  - Detailed Design
  - Module code and test
  - Integrate and test
- Cost of each subsystem is estimated separately. This reduces the margin of error.

3. DETAILED COCOMO MODEL

- Multiply all 15 Cost Drivers to get **Effort Adjustment Factor(EAF)**
- **E(Effort) = $a_b(KLOC)^{bb} \times EAF$**(in Person-Month)
- **D(Development Time) = $c_b(E)^{db}$** (in month)
- **E_p (Total Effort) = $\mu_p \times E$** (in Person-Month)
- **D_p (Total Development Time) = $\tau_p \times D$** (in month)

3. DETAILED COCOMO MODEL

Consider you are developing a full screen editor as a project. The major components identified and their sizes are

(i) Screen Edit – 4K
(ii) Command Lang Interpreter – 2K
(iii) File Input and Output – 1K
(iv) Cursor movement – 2K
(v) Screen Movement – 3K.

Assume the Required software reliability is high, product complexity is high, analyst capability is high & programming language experience is low. Use COCOMO model to estimate cost and time for different phases.

3. DETAILED COCOMO MODEL

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Size of modules : 4 + 2 + 1 + 2 + 3 = 13 KLOC **[Organic]**

Cost Drivers	Very Low	Low	Nominal	High	Very High	Extra High
RELY	0.75	0.88	1.00	1.15	1.40	--
CPLX	0.70	0.85	1.00	1.15	1.30	1.65
ACAP	1.46	1.19	1.00	0.86	0.71	
LEXP	1.14	1.07	1.00	0.95	--	--

EAF = 1.15 * 1.15 * 0.86 * 1.07 = 1.2169

3. DETAILED COCOMO MODEL

Initial Effort (E) = $a_b(KLOC)^{bb} \times EAF = 3.2\ (12)^{1.05} \times 1.2169$
= 52.9 person-months

Initial Development Time (D) = $c_b(E)^{d_b} = 2.5*(52.9)^{0.38} = 11.29$ months

Phase value of μ_p and τ_p

	Plan & Req	System Design	Detail Design	Module code & test	Integration & Test
Organic Small μ_p	0.06	0.16	0.26	0.42	0.16
Organic Small τ_p	0.10	0.19	0.24	0.39	0.18

3. DETAILED COCOMO MODEL

Phase wise effort & development time distribution

	E	D	Ep (in person-months)	Dp (in months)
Plan & Requirement	52.9	11.29	0.06*52.9 = 3.17	0.10*11.29=1.12
System Design				
Detail Design				
Module code & test				
Integration & test				

Reference: COCOMO-I, Classroom Presentation, Jagat Man Mulguthi, NCIT, Spring 2018